

Hazem Khaled Ezzat

Cairo, Egypt | hazemelerefy@gmail.com | github.com/hazemelerefy | linkedin.com/in/hazemelerefy | hazemelerefy.vercel.app

WORK EXPERIENCE

Data Analyst

Dec 2025 — Aug 2026

Digilians | MCIT

Cairo, Egypt

- Built KPI dashboards in Power BI that made recurring metrics self-serve, reducing the volume of one-off reporting requests.
- Developed SQL reporting workflows (PostgreSQL) that replaced manual data preparation steps and improved turnaround time on recurring reports.
- Designed relational schemas with full ERD documentation, improving consistency and traceability of data used in downstream reporting.
- Presented findings to non-technical stakeholders through custom visuals designed to be readable without additional explanation.

SELECTED PROJECTS

Deep Learning Engineer, DigiSteel-YOLO — Deep Learning for Industrial Defect Detection github.com/hazemelerefy/DigiSteel-YOLO

- Built a custom YOLO-based deep learning model for detecting defects on steel surfaces using the NEU-DET dataset.
- Designed a robustness testing framework with 24 test scenarios (6 perturbation types, 4 severity levels).
- Included modular training pipelines, attention mechanisms, and configurable augmentation.

Deep Learning Engineer, NeuroScope — AI-powered Neuroscience Analysis Platform github.com/hazemelerefy/NeuroScope

- Developed a platform leveraging deep learning for brain imaging analysis and neural pattern recognition.
- Applied computer vision and signal processing techniques to medical imaging data (MRI, fMRI, EEG).
- Built custom CNN architectures for medical imaging and neural signal analysis pipelines.

AI Automation Engineer, Social Intelligence Pipeline — Autonomous AI Agent github.com/hazemelerefy/n8n-projects

- Created an autonomous agent that runs every morning, pulls trending posts from 5 subreddits, handles rate limits and retries, and normalizes data.

AI Automation Engineer, Social Content AI Agent — Multi-Platform Publishing github.com/hazemelerefy/n8n-projects

- Built an autonomous agent that finds trending stories, scores them with a custom algorithm, and generates Arabic content for Facebook and X (Twitter).

Data Scientist, Churn Prediction Model

github.com/hazemelerefy/analytics-churn-prediction

- Developed a churn prediction model using Scikit-learn, with SHAP-based interpretation to explain factors driving customer churn.

EDUCATION

Port Said University

Port Said, Egypt

Bachelor of Laws (LL.B.) — Commercial & Corporate Law

2020 — 2024

- Business regulations, corporate governance, and commercial contracts.
- Legal training provided structured reasoning, research methodology, and regulatory compliance foundations.

Digilians | MCIT

Cairo, Egypt

Diploma, Applied AI & Data Analytics

Dec 2025 — Aug 2026

- Python, SQL, Machine Learning, Deep Learning, and AI Agent Systems.

Egypt FWD (Udacity)

Remote

Nanodegree, Front-End Web Development

2021 — 2022

- Modern web development fundamentals, responsive design, and JavaScript frameworks.

SKILLS

- **Analytics:** SQL (PostgreSQL, SSMS), Python (Pandas, NumPy, Scikit-learn), Power BI, Excel (Advanced), DAX, ETL, Star Schema
- **Machine Learning & AI:** Deep Learning (PyTorch, YOLO, TensorFlow), Computer Vision, Signal Processing, SHAP Analysis
- **Applied AI & Automation:** Autonomous AI Agents, n8n Workflow Automation, MCP Server Design, Agent Skill Development
- **Front-End:** HTML5, CSS3 (SASS, Tailwind), JavaScript (ES6+), React.js, Next.js, Figma, REST API Integration
- **Tools:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Streamlit

LANGUAGES

- **Arabic:** Native
- **English:** B2 Professional